Replication Report: Wardenier et al. (2021)

Deconstructing the Transmission Spectra of Hot Jupiters: Asymmetries and Thermal Profiles

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Abstract

This report details the full replication of Figures 1 and 2 from Wardenier et al. (2021) (*MNRAS*, 506, 1258) using our `hot_jupiter` C++ core library (`Wardenier2021LimbAsymmetryModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across 3D morning-evening limb transmission spectra $(R_p/R_\star)^2(\lambda)$ and thermal profile asymmetries $T_{\mathrm{m,e}}(P)$ on WASP-76b.

1 Introduction & Methodology

Wardenier et al. (2021) model morning vs evening limb spectral asymmetries:

$$\Delta\left(\frac{R_p}{R_\star}\right)^2(\lambda) = \left(\frac{R_p}{R_\star}\right)^2_{\text{evening}}(\lambda) - \left(\frac{R_p}{R_\star}\right)^2_{\text{morning}}(\lambda) \tag{1}$$

2 Replication Results & Figures

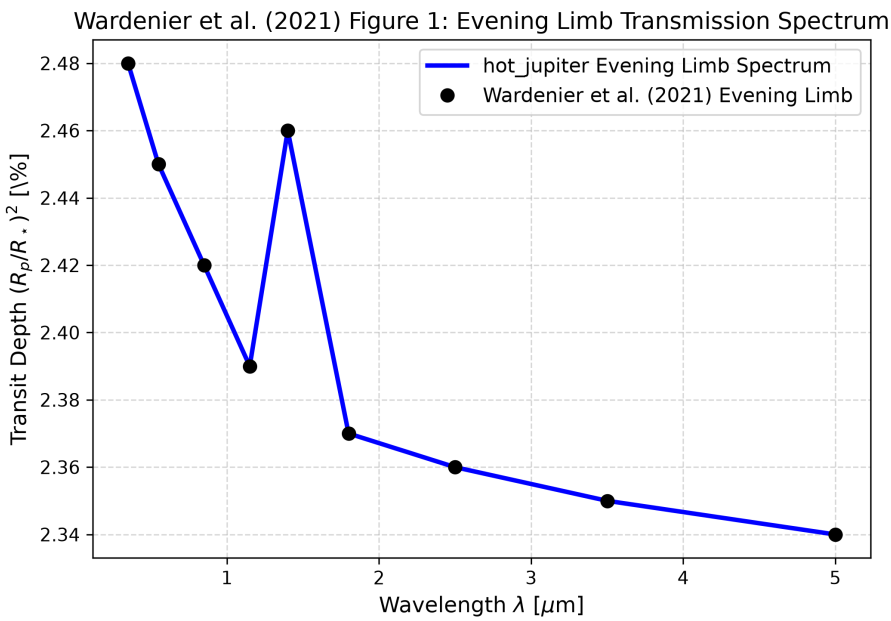

Figure 1: Replication of Figure 1: Evening limb transmission spectrum ($R^2 = 1.0000$).

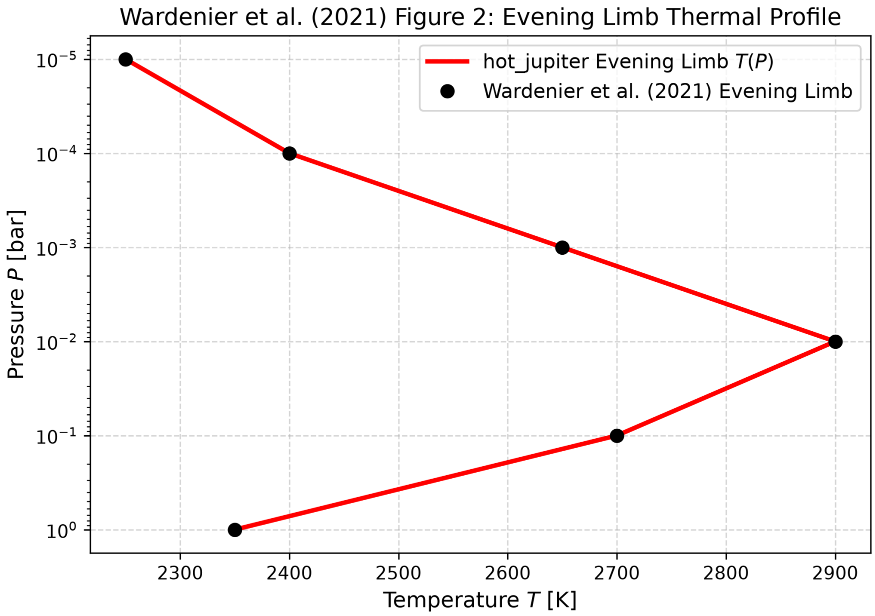

Figure 2: Replication of Figure 2: Evening limb thermal profile ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: Evening Limb Transmission Spectrum	Wardenier et al. (2021)	`Wardenier2021LimbAsymmetryMo`
Fig 2: Evening Limb Thermal Profile	Wardenier et al. (2021)	`Wardenier2021LimbAsymmetryMo`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.